Hojun Jeong

wjdghwns12@gachon.ac.kr | Personal Website | GitHub

RESEARCH INTERESTS

Knowledge Editing; Knowledge Retention and Forgetting; Sequential Model Updates; Representation Dynamics; Mechanistic Interpretability

EDUCATION

Gachon University — M.S. in AI Software (Artificial Intelligence) Mar. 2025 – Present
Advisor: Prof. Sangwoo Kang | GPA: 4.3 / 4.5

Gachon University — B.S. in AI Software (Artificial Intelligence) Mar. 2023 – Feb. 2025
GPA: 3.6 / 4.5

Dongguk University — Coursework in Computer Engineering Mar. 2018 – Feb. 2023
Transferred to Gachon University

PUBLICATIONS AND MANUSCRIPTS

Peer-Reviewed International Conference

***Hojun Jeong**, *Gyunyeop Kim, and Sangwoo Kang. **"KLOD: Locality-Preserving Knowledge Editing via Non-Target Distribution Preservation."** *Findings of the Association for Computational Linguistics: EMNLP 2026.* Accepted / to appear. [Paper] [Code]

Manuscripts Under Review

***Hojun Jeong**, *Gyunyeop Kim, and Sangwoo Kang. **"Probability-Lift Fine-Tuning for Knowledge Editing: Target-Token Prior as a Signal of Update Demand."** Under review at Expert Systems with Applications.

Hojun Jeong, and Sangwoo Kang. **"Beyond Norm Growth: A Geometric Decomposition of Hidden-State Drift in Sequential Knowledge Editing."** Under review at an MDPI journal.

** Equal contribution.*

Domestic Conference Publications

Hojun Jeong, and Sangwoo Kang. **"A Verify-Refine Hybrid for Dialogue Summarization with Topic Segmentation and Keypoint Coverage."** HCLT 2025, pp. 359–364, Oct. 2025. [In Korean]

Hojun Jeong, and Sangwoo Kang. **"Structured Zero-Shot Prompts through Rule-Based Intent Analysis for Dialogue Summarization."** Korea Computer Congress (KCC 2025), July 2025. [In Korean]

RESEARCH EXPERIENCE

Graduate Research Student, ISNLP Lab, Gachon University Mar. 2025 – Present

Advisor: Prof. Sangwoo Kang

KLOD — Locality-Preserving Sequential Knowledge Editing | EMNLP 2026 Findings

- Identified unintended drift in non-target output distributions under cross-entropy editing and designed bounded target logit-odds optimization with distribution-preserving KL constraints.
- Led implementation, baselines, experimental design, and ablations across two backbones and two datasets; evaluated sequential editing up to 150,000 edits.
- On LLaMA-3-8B-Instruct / CounterFact, improved locality from 1.87 to 44.75 while retaining 99.70 reliability; on zsRE, achieved 86.47 locality and 99.89 reliability.

RESEARCH EXPERIENCE (CONTINUED)

PLFT-KE — Target-Prior-Aware Fine-Tuning for Knowledge Editing | Under review

- Studied forgetting in a controlled synthetic biography setting (50K pre-training and 50K fine-tuning individuals); found target-token prior more informative than general output-token confidence in that setting.
- Developed a two-stage probability-lift initialization: optimize a latent shift with the model frozen, convert it into a rank-1 weight update, then consolidate with standard fine-tuning.
- On LLaMA-3.2-3B / CounterFact, improved FT-M score from 58.43 to 72.41 and specificity from 33.55 to 59.40; examined weaker and heterogeneous gains in long sequential editing.

Beyond Norm Growth — Geometric Analysis of Hidden-State Drift | Under review

- Decomposed Base-relative hidden-state displacement into parallel and orthogonal components, distinguishing representation drift from scalar norm growth across models, datasets, and editing conditions.
- Observed ~98.7–99.3% checkpoint-median orthogonal displacement-energy share at the measured boundary; tested controlled hidden-state interventions against same-final-norm parallel controls.
- For 34 eligible endpoint states, 25% orthogonal reduction produced mean changes of +0.729 pp locality and +0.333 pp teacher-forced efficacy, with heterogeneous effects across states.

Ongoing Research — Early-Stage Forgetting and Layer-Wise Transformations

- Investigating early fine-tuning forgetting and layer-wise rotational/scaling changes to connect representation subspaces, output distributions, and factual recall.

Research Assistant, NLP / AI Lab, Gachon University Sep. 2023 – Feb. 2025

Advisor: Prof. Sangwoo Kang

- Developed fine-tuning, prompting, and evaluation pipelines for dialogue summarization and knowledge editing; contributed to Korean conference publications.

AWARDS AND HONORS

Outstanding Presentation Paper Award, HCLT 2025 (Oct. 2025), for "A Verify-Refine Hybrid for Dialogue Summarization with Topic Segmentation and Keypoint Coverage."

PRESENTATIONS

HCLT 2025 — Oral presentation, "A Verify-Refine Hybrid for Dialogue Summarization with Topic Segmentation and Keypoint Coverage."

KCC 2025 — Poster presentation, "Structured Zero-Shot Prompts through Rule-Based Intent Analysis for Dialogue Summarization."

TECHNICAL SKILLS

Programming and Tools: Python, PyTorch, Hugging Face Transformers, EasyEdit, Git, Linux, LaTeX; NumPy, Pandas, Matplotlib.

Research Methods: Knowledge editing, localized fine-tuning, KL-based regularization, representation analysis, sequential evaluation, ablation studies.